

AN INTEGRAL COMPARISON OF SIGNATURE CLASSES IN K - AND L -THEORY

ABSTRACT. Let M be a closed oriented smooth manifold of dimension n . We prove that the comparison map from connective complex K -theory to the L -theory of $\mathbb{C}$ sends the signature class of M to $2^{\lfloor n/2 \rfloor}$ times its Sullivan–Ranicki fundamental class, integrally. This answers Problem 8.4 of Land, Nikolaus, and Schlichting. The proof lifts both classes to the Thom spectrum of the virtual bundle $\varepsilon^n - \gamma_n$ over $\mathrm{BSO}(n)$. Ebert’s universal signature symbol determines their rational comparison. At the prime 2, the remaining difference is detected by rationalization and reduction modulo 2, using the integral cohomology of $\mathrm{BSO}(n)$ and the Eilenberg–MacLane decomposition of connective L -theory. The result gives an integral comparison for classes mapped to an arbitrary space, a specified divisibility statement, and a corresponding equality of assembled signature classes.

1. INTRODUCTION

The signature operator and the Sullivan–Ranicki orientation associate signature classes to an oriented manifold in complex K -homology and symmetric L -homology, respectively. Land, Nikolaus, and Schlichting constructed a multiplicative comparison between these theories and computed its effect on signature classes after inverting 2. Their Problem 8.4 asks whether the resulting equality holds integrally [5, Proposition 8.3 and Problem 8.4].

Write $\ell(\mathbb{C})$ for the connective cover of the projective symmetric L -theory spectrum $L(\mathbb{C})$, where $\mathbb{C}$ carries complex conjugation. The comparison and orientation maps are

$$\tau_{\mathbb{C}} : \mathrm{ku} \longrightarrow \ell(\mathbb{C}), \quad \sigma_{\mathbb{C}} : \mathrm{MSO} \longrightarrow \ell(\mathbb{C}).$$

Here $\sigma_{\mathbb{C}}$ is the Sullivan–Ranicki orientation followed by extension of scalars from $\mathbb{R}$ to $\mathbb{C}$. For a closed oriented smooth n -manifold M , let $[M] \in \mathrm{MSO}_n(M)$ be the bordism class of its identity map. We use the signature class $[D_M] \in \mathrm{ku}_n(M)$ normalized as in [5, Definition 8.2]: the analytic class in $\mathrm{KU}_0(M)$ or $\mathrm{KU}_1(M)$ is multiplied by $\beta^{\lfloor n/2 \rfloor}$, where $\beta \in \pi_2 \mathrm{ku}$ is the complex Bott element.

Theorem 1.1. *For every closed oriented smooth manifold M of dimension $n \geq 0$,*

$$(1.1) \quad \tau_{\mathbb{C}}([D_M]) = 2^{\lfloor n/2 \rfloor} \sigma_{\mathbb{C}}([M]) \quad \text{in } \ell(\mathbb{C})_n(M).$$

The comparison in [5, Proposition 8.3] says that the difference in (1.1) is killed by a power of 2. Theorem 1.1 removes this possible torsion error. In particular, for every continuous map $f : M \rightarrow X$ it identifies $\tau_{\mathbb{C}}(f_*[D_M])$ with $2^{\lfloor n/2 \rfloor} \sigma_{\mathbb{C}}([M, f])$. Thus the image of the analytic class has a specified integral preimage under multiplication by $2^{\lfloor n/2 \rfloor}$, namely the topological signature class. Applied to $X = BG$, this equality also determines the comparison after assembly; the precise statement is given in Corollary 6.2.

The proof takes place first on a Thom spectrum over $\mathrm{BSO}(n)$. The universal signature symbol gives a connective K -class there, while the Sullivan–Ranicki orientation gives an L -class. Their rational equality follows from the characteristic series of the signature symbol. The identity $\tau_{\mathbb{C}}(\beta) = 2b$, with $b \in \pi_2 \ell(\mathbb{C})$ the periodicity generator, implies that the comparison modulo 2 factors through ordinary integral cohomology. The degree-zero component of the universal signature class is $2^{\lfloor n/2 \rfloor}$ times the ordinary Thom class, so the universal difference also vanishes modulo 2 when $n \geq 2$. Finally, the 2-local L -cohomology of this Thom spectrum is a product of ordinary cohomology groups of $\mathrm{BSO}(n)$. Their

Key words and phrases. signature operator, connective K -theory, L -theory, Thom spectrum.

torsion is killed by 2, which makes rationalization and reduction modulo 2 jointly injective. Pullback to M and the comparison away from 2 finish the proof.

Conventions. For a spectrum E , we use $E^q(X) = [X, \Sigma^q E]$ and $E_q(Y) = \pi_q(E \wedge \Sigma^\infty Y_+)$. In cohomological notation β has degree -2 . A subscript (2) denotes localization at the prime 2, and $E_{\mathbb{Q}} = E \wedge H\mathbb{Q}$. The notation $x_{\mathbb{Q}}$ always denotes the image under coefficient rationalization. The spectrum $E/2$ is the cofiber of multiplication by 2 on E . All K -theory is complex unless otherwise specified. A manifold is smooth, closed, and oriented unless a different hypothesis is stated.

2. SIGNATURE SYMBOLS ON THE UNIVERSAL THOM SPECTRUM

Let γ_n be the universal oriented real n -plane bundle over $\text{BSO}(n)$. Define

$$(2.1) \quad T_n = \text{Th}(\varepsilon^n - \gamma_n) = \Sigma^n \text{MTSO}(n).$$

The virtual bundle has rank zero, so T_n is connective. Its ordinary Thom class $U_n \in H^0(T_n; \mathbb{Z})$ gives, for each coefficient group A ,

$$(2.2) \quad H^q(T_n; A) \cong H^q(\text{BSO}(n); A).$$

We use the Thom diagonal to interpret expressions $U_n x$ with $x \in H^*(\text{BSO}(n); A)$.

The oriented virtual bundle $\varepsilon^n - \gamma_n$ determines a map

$$j_n : T_n \longrightarrow \text{MSO}.$$

It also determines the class

$$(2.3) \quad v_n = j_n^* \sigma_{\mathbb{C}} \in \ell(\mathbb{C})^0(T_n).$$

If M has dimension n , its tangent bundle classifying map induces

$$(2.4) \quad c_M : \Sigma^n M^{-TM} \longrightarrow T_n.$$

Here M^{-TM} denotes the Thom spectrum of the negative tangent bundle. Atiyah duality gives, naturally in the coefficient spectrum E ,

$$(2.5) \quad E^0(\Sigma^n M^{-TM}) \cong E_n(M).$$

Under this identification $c_M^* v_n$ corresponds to $\sigma_{\mathbb{C}}([M])$: the normal Thom representative of the bordism fundamental class is the pullback of j_n , and applying the orientation gives the asserted equality.

2.1. The analytic normalization. Choose a Riemannian metric on M and put $D = d + d^\dagger$ on complex differential forms. We distinguish the adjoint $d^\dagger$ from the Hodge operator $*$. The chirality involution on p -forms is

$$(2.6) \quad \tau\omega = i^{[n/2] + p(p+1) + 2p(n-p)} * \omega.$$

For even n , the grading by τ gives an analytic class in $\text{KU}_0(M)$. For odd n , D commutes with τ ; its restriction to the $+1$ eigenspace of τ gives the analytic class in $\text{KU}_1(M)$. This is the Clifford description used in [5, §8, Definition 8.2]; the eigenspace description is explained in [6, §1]. Writing

$$m = \lfloor n/2 \rfloor, \quad \epsilon = n - 2m \in \{0, 1\},$$

and denoting this analytic class by $[D_M]_{\text{an}}$, the normalization is

$$(2.7) \quad [D_M] = \beta^m [D_M]_{\text{an}} \in \text{KU}_n(M) \cong \text{ku}_n(M).$$

The last isomorphism follows from the dimension of M , or from (2.5) and the connectivity of $\Sigma^n M^{-TM}$.

Proposition 2.1. *For every $n \geq 0$ there is a class $a_n \in \mathrm{ku}^0(T_n)$ such that*

$$c_M^* a_n \longleftrightarrow [D_M]$$

under (2.5), for every n -manifold M .

Proof. The signature principal symbol is $\mathrm{SO}(n)$ -equivariant. Applying the associated-bundle construction to the universal oriented frame bundle gives its universal symbol class. Ebert constructs this class and its image on $\mathrm{MTSO}(n)$ using the relative complex K -Thom isomorphism [3, §3.2]. Choose the integer grading so that the resulting classes are

$$s_{2r} \in \mathrm{KU}^0(\mathrm{MTSO}(2r)), \quad s_{2r-1} \in \mathrm{KU}^{-1}(\mathrm{MTSO}(2r-1)).$$

For the odd symbol, this convention is the suspension convention obtained from the symbol $\mathrm{smb}(D) - it$ on the tangent bundle with an additional real coordinate. On a manifold, this is a class in $\mathrm{KU}_c^0(TM \oplus \mathbb{R}) \cong \mathrm{KU}_c^{-1}(TM)$.

More explicitly, the relative Thom map for the negative complexification of a real rank- n bundle lowers the integer cohomological degree by $2n$. Multiplication by β^{-n} makes this a degree-preserving periodic Thom map. Applied to the symbol of degree $-\epsilon$, it gives the class s_n of degree $-\epsilon$ just specified. Naturality of this construction identifies the pullback of s_n to M^{-TM} with the periodic relative Thom image of the principal symbol of D_M .

The composite

$$\mathrm{KU}_c^{-\epsilon}(TM) \xrightarrow{\beta^{-n} \mathrm{th}_{-TM\mathbb{C}}} \mathrm{KU}^{-\epsilon}(M^{-TM}) \xrightarrow{\text{Atiyah duality}} \mathrm{KU}_\epsilon(M)$$

sends the principal symbol to $[D_M]_{\mathrm{an}}$, where $TM_{\mathbb{C}} = TM \otimes_{\mathbb{R}} \mathbb{C}$. This identification is the K -homology form of the index theorem; see [1, Theorem 4.3] and its all-parities KK -theoretic formulation [4, Theorem 4.2 and Remark 4.7]. It is an equality of classes on M , before pushforward to a point.

For completeness, the odd operator in this construction has the same normalization as (2.6). If $n = 2r - 1$ and $\phi \in \Omega^{2q}(M; \mathbb{C})$, then

$$\tau\phi = i^r(-1)^q * \phi.$$

Let E_+ be the $+1$ eigenspace of τ . The map

$$U : \Omega^{\mathrm{ev}}(M; \mathbb{C}) \longrightarrow E_+, \quad U\phi = 2^{-1/2}(\phi + \tau\phi)$$

is unitary, and $U^{-1}(D|_{E_+})U = \tau D|_{\Omega^{\mathrm{ev}}}$. Since $d^\dagger\phi = *d*\phi$ on even forms in odd dimension, one obtains

$$(2.8) \quad \tau D\phi = i^r(-1)^{q+1}(*d - d*)\phi.$$

This is Ebert's odd signature operator [3, §4.1]. In even dimension both constructions use the grading $i^{r+p(p-1)}*$ on p -forms.

Now $\beta^m s_n$ has degree $-n$, and hence defines

$$\tilde{a}_n \in \mathrm{KU}^{-n}(\mathrm{MTSO}(n)) = \mathrm{KU}^0(T_n).$$

Because T_n is connective, the connective-cover map induces an isomorphism $[T_n, \mathrm{ku}] \rightarrow [T_n, \mathrm{KU}]$. Let a_n be the unique lift of $\tilde{a}_n$. Naturality of (2.5) and (2.7) gives the required pullback identity. $\square$

3. RATIONAL NORMALIZATION AND THE ORDINARY THOM COMPONENT

Land, Nikolaus, and Schlichting construct an orientation $\mathcal{L}_{\mathrm{AS}} : \mathrm{MSO} \rightarrow \mathrm{ko}[1/2]$. Let

$$(3.1) \quad \lambda_n = j_n^*(c\mathcal{L}_{\mathrm{AS}}) \in \mathrm{ku}[1/2]^0(T_n),$$

where c is complexification. Its Hirzebruch characteristic series is

$$(3.2) \quad Q_{\mathrm{AS}}(z) = \frac{z/2}{\tanh(z/2)}.$$

Moreover, the two orientations agree after applying the comparison:

$$(3.3) \quad \tau_{\mathbb{C}} \circ c\mathcal{L}_{\text{AS}} = \sigma_{\mathbb{C}}[1/2].$$

These are [5, Theorem 8.5(3)–(4)] and the proof of Proposition 8.3].

Proposition 3.1. *For every $n \geq 0$,*

$$(3.4) \quad (a_n)_{\mathbb{Q}} = 2^{\lfloor n/2 \rfloor} (\lambda_n)_{\mathbb{Q}} \quad \text{in } \text{ku}_{\mathbb{Q}}^0(T_n).$$

Proof. For an oriented real bundle V , write $\mathcal{L}_{4k}^{\text{AS}}(V)$ for the degree- $4k$ component of the multiplicative characteristic class associated with (3.2). Thus $\mathcal{L}_0^{\text{AS}}(V) = 1$, and in terms of the usual Hirzebruch classes, $\mathcal{L}_{4k}^{\text{AS}}(V) = 2^{-2k} L_k(V)$.

Under the rational Chern character, an orientation with characteristic series Q has Thom class $u_H(V)Q(V)^{-1}$. For the virtual bundle $V = \varepsilon^n - \gamma_n$, this inverse is $Q(\gamma_n)$. Consequently the components of $\text{ch}((\lambda_n)_{\mathbb{Q}})$ are

$$(3.5) \quad \text{ch}_{4k}((\lambda_n)_{\mathbb{Q}}) = U_n \mathcal{L}_{4k}^{\text{AS}}(\gamma_n), \quad \text{ch}_{4k+2}((\lambda_n)_{\mathbb{Q}}) = 0.$$

Here ch_{2j} is the component in $H^{2j}(T_n; \mathbb{Q})$ indexed by β^j in the rational splitting of ku .

For $n = 2r$, Ebert computes the universal signature class by the series

$$z \coth(z/2) = 2Q_{\text{AS}}(z)$$

[3, Proposition 4.3.4(2)]. Multiplying over the r formal roots and making the degree shift in Proposition 2.1 therefore gives

$$(3.6) \quad \text{ch}_{4k}((a_{2r})_{\mathbb{Q}}) = 2^r U_{2r} \mathcal{L}_{4k}^{\text{AS}}(\gamma_{2r}), \quad \text{ch}_{4k+2}((a_{2r})_{\mathbb{Q}}) = 0.$$

For odd dimension, the inclusion $\text{BSO}(2r-1) \rightarrow \text{BSO}(2r)$ induces

$$\iota_r : \Sigma^{-1} \text{MTSO}(2r-1) \longrightarrow \text{MTSO}(2r), \quad t_r = \Sigma^{2r} \iota_r : T_{2r-1} \longrightarrow T_{2r}.$$

The signature restriction formula of [3, Proposition 4.3.4(1) and diagram (4.3.2)], with the integer grading restored, is

$$(3.7) \quad \iota_r^* s_{2r} = 2\beta^{-1} s_{2r-1} \quad \text{in } \text{KU}^1(\text{MTSO}(2r-1)).$$

Indeed the left side has degree 1 on $\text{MTSO}(2r-1)$, whereas s_{2r-1} was placed in degree -1 ; multiplication by β^{-1} is the corresponding periodicity identification. It follows that

$$(3.8) \quad t_r^* a_{2r} = 2a_{2r-1}.$$

The ordinary Thom classes pull back compatibly, and γ_{2r} restricts to $\gamma_{2r-1} \oplus \mathbb{R}$. Since $Q_{\text{AS}}(0) = 1$, pulling back (3.6) and using (3.8) gives

$$\text{ch}_{4k}((a_{2r-1})_{\mathbb{Q}}) = 2^{r-1} U_{2r-1} \mathcal{L}_{4k}^{\text{AS}}(\gamma_{2r-1}), \quad \text{ch}_{4k+2}((a_{2r-1})_{\mathbb{Q}}) = 0.$$

These component equalities determine the rational class. In fact,

$$\text{ku}_{\mathbb{Q}} \simeq \bigvee_{j \geq 0} \Sigma^{2j} H\mathbb{Q} \simeq \prod_{j \geq 0} \Sigma^{2j} H\mathbb{Q},$$

so that $\text{ku}_{\mathbb{Q}}^0(T_n) \cong \prod_{j \geq 0} H^{2j}(T_n; \mathbb{Q})$. Together with (3.5), the formulas prove (3.4). The case $n = 0$ is the unit class. $\square$

Let $r : \text{ku}_{(2)} \rightarrow H\mathbb{Z}_{(2)}$ be the degree-zero Postnikov map.

Corollary 3.2. *For $m = \lfloor n/2 \rfloor$,*

$$(3.9) \quad r((a_n)_{(2)}) = 2^m U_n \quad \text{in } H^0(T_n; \mathbb{Z}_{(2)}).$$

Proof. The degree-zero component in Proposition 3.1 is $2^m U_n$. By (2.2), $H^0(T_n; \mathbb{Z}_{(2)}) \cong \mathbb{Z}_{(2)}$, and its map to rational cohomology is injective. This proves the integral equality at the prime 2. $\square$

4. DETECTION AT THE PRIME 2

Let $q : \ell(\mathbb{C})_{(2)} \rightarrow \ell(\mathbb{C})_{(2)}/2$ be the cofiber map. The two lemmas in this section show that the universal comparison is determined by its rational image and its degree-zero ordinary component.

Lemma 4.1. *For every $n \geq 0$, the map*

$$(4.1) \quad \ell(\mathbb{C})_{(2)}^0(T_n) \longrightarrow \ell(\mathbb{C})_{\mathbb{Q}}^0(T_n) \times (\ell(\mathbb{C})_{(2)}/2)^0(T_n), \quad x \longmapsto (x_{\mathbb{Q}}, qx)$$

is injective.

Proof. At the prime 2, $\ell(\mathbb{C})$ is the free $E_1\text{-}H\mathbb{Z}$ -algebra on a generator of degree 2; see [5, §8, the paragraph following Question 8.8]. Its underlying $H\mathbb{Z}_{(2)}$ -module therefore has a splitting

$$(4.2) \quad \ell(\mathbb{C})_{(2)} \simeq \bigvee_{j \geq 0} \Sigma^{2j} H\mathbb{Z}_{(2)} \simeq \prod_{j \geq 0} \Sigma^{2j} H\mathbb{Z}_{(2)}.$$

The map from the wedge to the product is an equivalence because its map on every homotopy group is an isomorphism. Rationalizing the wedge splitting and then using the analogous product splitting over $\mathbb{Q}$ gives a compatible decomposition of the coefficient rationalization map. Together with (2.2), this identifies the map on degree-zero cohomology with

$$(4.3) \quad \prod_{j \geq 0} H^{2j}(\mathrm{BSO}(n); \mathbb{Z}_{(2)}) \longrightarrow \prod_{j \geq 0} H^{2j}(\mathrm{BSO}(n); \mathbb{Q}).$$

Each ordinary cohomology group is finitely generated in its degree. The integral computation of $H^*(\mathrm{BSO}(n); \mathbb{Z})$ shows that its torsion is killed by 2 [2, Theorem 1.5, proof on pp. 285–286]. It follows that

$$(4.4) \quad 2 \ker(\ell(\mathbb{C})_{(2)}^0(T_n) \longrightarrow \ell(\mathbb{C})_{\mathbb{Q}}^0(T_n)) = 0.$$

Suppose $x_{\mathbb{Q}} = 0$ and $qx = 0$. Exactness for the cofiber sequence $\ell(\mathbb{C})_{(2)} \xrightarrow{2} \ell(\mathbb{C})_{(2)} \xrightarrow{q} \ell(\mathbb{C})_{(2)}/2$ gives $x = 2y$ for some $y \in \ell(\mathbb{C})_{(2)}^0(T_n)$. Rationally, $2y_{\mathbb{Q}} = 0$, so $y_{\mathbb{Q}} = 0$. Equation (4.4) then gives $x = 2y = 0$. $\square$

Remark 4.2. The argument uses coefficient rationalization and the compatible splittings of spectra in (4.2). For the infinite Thom spectrum T_n , it does not identify $\ell(\mathbb{C})_{(2)}^0(T_n) \otimes \mathbb{Q}$ with $\ell(\mathbb{C})_{\mathbb{Q}}^0(T_n)$; such an identification would require an additional argument concerning infinite products.

Lemma 4.3. *The composite*

$$\mathrm{ku}_{(2)} \xrightarrow{\tau_{\mathbb{C}}} \ell(\mathbb{C})_{(2)} \xrightarrow{q} \ell(\mathbb{C})_{(2)}/2$$

factors, in the homotopy category of spectra, through $r : \mathrm{ku}_{(2)} \rightarrow H\mathbb{Z}_{(2)}$.

Proof. The comparison is multiplicative and satisfies

$$(4.5) \quad \tau_{\mathbb{C}}(\beta) = 2b,$$

where $b \in \pi_2 \ell(\mathbb{C})$ is the periodicity generator [5, Remark 8.9]. The $H\mathbb{Z}_{(2)}$ -module structure gives

$$\ell(\mathbb{C})_{(2)}/2 \simeq \ell(\mathbb{C})_{(2)} \otimes_{H\mathbb{Z}_{(2)}} H\mathbb{F}_2.$$

Thus multiplication by 2 is nullhomotopic on $\ell(\mathbb{C})_{(2)}/2$. If $m_b : \Sigma^2 \ell(\mathbb{C})_{(2)} \rightarrow \ell(\mathbb{C})_{(2)}$ is multiplication by b , multiplicativity and (4.5) imply

$$q\tau_{\mathbb{C}}\beta = q m_{2b} \Sigma^2 \tau_{\mathbb{C}} = 2(q m_b \Sigma^2 \tau_{\mathbb{C}}) = 0.$$

The cofiber sequence

$$\Sigma^2 \mathrm{ku}_{(2)} \xrightarrow{\beta} \mathrm{ku}_{(2)} \xrightarrow{r} H\mathbb{Z}_{(2)}$$

therefore supplies a map $h : H\mathbb{Z}_{(2)} \rightarrow \ell(\mathbb{C})_{(2)}/2$ with $q\tau_{\mathbb{C}} = hr$ in the homotopy category. $\square$

5. THE INTEGRAL COMPARISON

We first prove the equality on the universal Thom spectrum.

Theorem 5.1. *For every $n \geq 0$, the classes of (2.3) and Proposition 2.1 satisfy*

$$(5.1) \quad \tau_{\mathbb{C}}((a_n)_{(2)}) = 2^{\lfloor n/2 \rfloor} (v_n)_{(2)} \quad \text{in } \ell(\mathbb{C})_{(2)}^0(T_n).$$

Proof. Let $m = \lfloor n/2 \rfloor$ and consider the difference

$$\delta_n = \tau_{\mathbb{C}}((a_n)_{(2)}) - 2^m (v_n)_{(2)}.$$

Proposition 3.1 and (3.3) imply

$$(\delta_n)_{\mathbb{Q}} = 2^m (\tau_{\mathbb{C}}((\lambda_n)_{\mathbb{Q}}) - (v_n)_{\mathbb{Q}}) = 0.$$

For $n \geq 2$, Corollary 3.2 and Lemma 4.3 give

$$q\tau_{\mathbb{C}}((a_n)_{(2)}) = h(r((a_n)_{(2)})) = h(2^m U_n) = 0.$$

The last equality holds because $m \geq 1$ and $[T_n, \ell(\mathbb{C})_{(2)}/2]$ is killed by 2. Also $q(2^m (v_n)_{(2)}) = 0$, so $q\delta_n = 0$. Lemma 4.1 now gives $\delta_n = 0$.

If $n = 0$ or 1, then $\text{BSO}(n)$ is a point and $T_n \simeq S^0$. In these cases $\ell(\mathbb{C})_{(2)}^0(T_n) = \mathbb{Z}_{(2)}$ is torsion-free, and the rational equality already gives $\delta_n = 0$. $\square$

Proof of Theorem 1.1. Pull back (5.1) along c_M and apply (2.5). Proposition 2.1 identifies the K -class with $[D_M]$, and (2.3) identifies the L -class with $\sigma_{\mathbb{C}}([M])$. The naturality of duality in the coefficient spectrum gives (1.1) after localization at 2.

Let d_M be the integral difference between the two sides of (1.1). By [5, Proposition 8.3], there is an integer $N \geq 0$ with $2^N d_M = 0$. Since d_M vanishes after localization at 2, some odd integer s satisfies $sd_M = 0$. Choose integers u, v with $us + v2^N = 1$. Then $d_M = usd_M + v2^N d_M = 0$. $\square$

6. CONSEQUENCES FOR SIGNATURE CLASSES

6.1. Classes over a space and integral divisibility.

Corollary 6.1. *Let X be a space and $f : M \rightarrow X$ a continuous map from an n -manifold. Then*

$$(6.1) \quad \tau_{\mathbb{C}}(f_*[D_M]) = 2^{\lfloor n/2 \rfloor} \sigma_{\mathbb{C}}([M, f]) \quad \text{in } \ell(\mathbb{C})_n(X),$$

where $[M, f] \in \text{MSO}_n(X)$ is the oriented bordism class of the map. In particular,

$$\tau_{\mathbb{C}}(f_*[D_M]) \in 2^{\lfloor n/2 \rfloor} \ell(\mathbb{C})_n(X).$$

Proof. Apply f_* to Theorem 1.1 and use the naturality of $\tau_{\mathbb{C}}$ and $\sigma_{\mathbb{C}}$. $\square$

The corollary specifies a preimage under multiplication by $2^{\lfloor n/2 \rfloor}$: it is the class $\sigma_{\mathbb{C}}([M, f])$. Other preimages differ from it by an element annihilated by this power of 2. The theorem therefore supplies a specified topological preimage of the analytic comparison, even when multiplication by 2 is not injective. For dimensions 2, 3 the divisibility factor is 2; for dimensions 4, 5 it is 4.

There is also a necessary condition for a class in $\text{ku}_n(X)$ to admit a signature representative. Put $d = 2^{\lfloor n/2 \rfloor}$ and let $q_d : \ell(\mathbb{C}) \rightarrow \ell(\mathbb{C})/d$ be the cofiber map for multiplication by d . Exactness gives

$$(q_d)_* \tau_{\mathbb{C}}(f_*[D_M]) = 0.$$

Consequently, if $(q_d)_* \tau_{\mathbb{C}}(\alpha) \neq 0$ for $\alpha \in \text{ku}_n(X)$, then α cannot equal $f_*[D_M]$ for any closed oriented n -manifold mapped to X .

As a further direct consequence, if $f : M^n \rightarrow X$ and $g : N^n \rightarrow X$ satisfy $f_*[D_M] = g_*[D_N]$ in $\mathrm{ku}_n(X)$, then

$$2^{\lfloor n/2 \rfloor} (\sigma_{\mathbb{C}}([M, f]) - \sigma_{\mathbb{C}}([N, g])) = 0.$$

When $\ell(\mathbb{C})_n(X)$ has no 2-primary torsion, the two topological signature classes are consequently equal.

6.2. Assembly of signature classes. Let G be a torsion-free discrete group, and let $C_r^*(G; \mathbb{C})$ be its reduced complex group C^* -algebra. Write

$$\mu_G : \mathrm{KU}_n(BG) \longrightarrow K_n(C_r^*(G; \mathbb{C}))$$

for the Baum–Connes assembly map. On the L -theory side let

$$\alpha_G : H_n(BG; L(\mathbb{C})) \longrightarrow L_n(\mathbb{C}G)$$

be projective symmetric L -theory assembly. The involution on $\mathbb{C}G$ is $(\sum z_g g)^* = \sum \overline{z_g} g^{-1}$. Let $i : \ell(\mathbb{C}) \rightarrow L(\mathbb{C})$ be the connective-cover map and $j : \mathbb{C}G \rightarrow C_r^*(G; \mathbb{C})$ the canonical homomorphism.

For $f : M^n \rightarrow BG$, define the degree- n analytic signature index by

$$\mathrm{Ind}_n(D_M, f) = \mu_G(f_*[D_M]),$$

where the argument of μ_G is understood via $\mathrm{ku} \rightarrow \mathrm{KU}$.

Corollary 6.2. *For $n \geq 0$ and $f : M^n \rightarrow BG$ as above, the following equality holds in $L_n(C_r^*(G; \mathbb{C}))$:*

$$(6.2) \quad \tau_{C_r^*(G; \mathbb{C})}(\mathrm{Ind}_n(D_M, f)) = 2^{\lfloor n/2 \rfloor} L_n(j) \alpha_G i_*(\sigma_{\mathbb{C}}([M, f])).$$

Proof. The orbit-category argument of [5, Theorem 7.4] applies to complex group algebras as well. Following it by the map from the full to the reduced group C^* -algebra, as in the proof of [5, Theorem 7.1], gives

$$\tau_{C_r^*(G; \mathbb{C})} \mu_G = L_n(j) \alpha_G i_* \tau_{\mathbb{C}}$$

on $\mathrm{ku}_n(BG)$. Substitute (6.1). Since $n \geq 0$, the degree- n group of connective K -theory of the coefficient algebra agrees with its periodic K -group. $\square$

The assembly comparison itself is an input from [5]. Corollary 6.2 specifies its value on geometric signature classes without a 2-primary error term. It requires no injectivity or surjectivity assumption on either assembly map. The torsion-free hypothesis identifies the domain of Baum–Connes assembly with $\mathrm{KU}_n(BG)$; for a group with torsion the corresponding statement concerns assembly from the free family.

6.3. The numerical specialization. For the map $p : M^{4d} \rightarrow \mathrm{pt}$, the analytic index formula gives

$$p_*[D_M] = \beta^{2d} \mathrm{sign}(M).$$

Thus (4.5) and Corollary 6.1 specialize to

$$\tau_{\mathbb{C}}(p_*[D_M]) = 2^{2d} b^{2d} \mathrm{sign}(M) = 2^{2d} \sigma_{\mathbb{C}}([M, p]).$$

This equality of numerical indices is already determined by the coefficient comparison. Theorem 1.1 retains the fundamental class on M , and Corollary 6.1 retains its image on an arbitrary X . Their additional assertion is the vanishing of the possible 2-primary difference in these homology groups.

The dimension dependence is essential to the formulation. The construction uses the individual spectra T_n and the relation $t_r^* a_{2r} = 2a_{2r-1}$. It is compatible with the factor of 2 in the boundary behavior of the odd signature class discussed in [6, Remark 4]. The normalization of the theorem concerns signature classes in each dimension, while the stable Atiyah–Singer orientation $\mathcal{L}_{\mathrm{AS}}$ used in the proof takes values in $\mathrm{ko}[1/2]$.

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